

Advanced (Carolina Reaper)

Q1. What is the value of $2 \times 3 + 11$?

- (A) 15
- (B) 17
- (C) 20
- (D) 21
- (E) 25

Q2. Simplify the expression $5(2x - 3) + 2(x + 1)$

- (A) $10x - 13$
- (B) $7x - 11$
- (C) $7x + 1$
- (D) $10x + 2$
- (E) $7x - 13$

Q3. Evaluate the expression $(18 - 3) \cdot (7 - 2)$

- (A) 40
- (B) 50
- (C) 60
- (D) 70
- (E) 75

Q4. What is the value of x in the equation $2x + 5 = 11$?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q5. A bookstore is having a sale where all books are 15

- (A) 20.25
- (A) 21.25
- (A) 22.25
- (A) 23.25
- (A) 24.25

Q6. What is the simplified form of $3(2x + 1) + 2(4x - 2)$?

- (A) $10x + 1$
- (B) $10x - 1$
- (C) $10x + 3$
- (D) $10x - 3$
- (E) $14x - 1$

Q7. A bus travels from City A to City B at an average speed of 40 miles per hour. If the distance between the cities is 160 miles, how many hours will the trip take?

- (A) 3 hours
- (B) 4 hours
- (C) 5 hours
- (D) 6 hours
- (E) 7 hours

Q8. Simplify the expression $(9 - 2) \cdot (11 + 2)$

- (A) 70
- (B) 77
- (C) 91
- (D) 112
- (E) 121

Open Ended Questions (FRQ)

Q9. Tom has been saving money for a new bike and has 120 in his savings account. He wants to buy a bike that costs 180. Tom's parents have agreed to give him an allowance of 5 per week for doing chores. How many weeks will it take Tom to have enough money to buy the bike? Show your work and explain your reasoning.

Q10. A water tank can hold 1200 gallons of water. Due to a leak, the tank is losing water at a rate of 5 gallons per hour. If 300 gallons of water are already in the tank, how many hours will it take for the tank to be empty? Show your work and explain your reasoning.

Chef's Tips

- Provide explicit instructions for each question
- Allow students to use calculators for certain questions
- Encourage students to show their work and explain their reasoning